Junpyo Jang

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RESEARCH INTERESTS

2D quantum materials, moiré magnetism, electron microscopy, spintronics, and beyond-CMOS devices

EDUCATION

Korea University, Seoul, South Korea — Mar 2017 – Feb 2023
B.S. in Materials Science and Engineering | GPA: 3.95/4.5 (Major: 4.27/4.5)

RESEARCH EXPERIENCE

Research Intern (Full-time) — Apr 2025 – Present
Quantum-material Spintronics Group (Advisor: Jun Woo Choi)
Center for Semiconductor Technology, Korea Institute of Science and Technology (KIST), Seoul, South Korea

Project 1 — Moiré-modulated magnetic textures in Fe_3GaTe_2

- Investigate how local moiré geometry, consistent with interlayer twist and strain, relates to zero-field-cooled stripe domains and field-cooled skyrmion lattices in Fe_3GaTe_2 using TEM/LTEM.
- Prepare TEM-grid specimens, conduct MOKE measurements, and plan TEM/LTEM experiments; established a within-flake comparison protocol to reduce sample variability.
- Developed a PyLorentz-based workflow for phase reconstruction and quantitative analysis of magnetic domains and skyrmion lattices; compared results with manual measurements.
- Developed a Python tool for local moiré mapping (period, orientation, model-based twist/strain), validated using synthetic TEM images from a custom JavaScript simulator.

Project 2 — Li-ion intercalation in vdW magnets by solid-state ionic gating (contributor)

- Prepare FIB specimens and participate in in situ TEM studies of Li-ion gating in Fe_3GaTe_2 and CrSBr.
- Use EELS to investigate Li incorporation and TEM/SAED to assess changes in interlayer spacing.

Project 3 — Strain-tunable magnetism by polymer-based tensile strain (contributor)

- Fabricate samples on polymer substrates and characterize their magnetic response to tensile strain using MOKE.

Project 4 — vdW heterostructure devices

- Assembled $Fe_{2.7}GeTe_2$/hBN/Fe_3GeTe_2 magnetic tunnel junction stacks using glovebox exfoliation and dry transfer.
- Fabricated a graphite/In_2Se_3/Fe_3GaTe_2/hBN heterostructure to investigate ferroelectric control of magnetism.

PUBLICATIONS & PRESENTATIONS

Manuscript in preparation

[1] Manuscript on moiré-modulated stripe domains in Fe_3GaTe_2 (co-first authors: **J. Jang** and M. Jang, author order pending).

Poster presentations

[1] **J. Jang**, M. Jang, H. Suh, K. Kim, and J. W. Choi, "Moiré-Modulated Magnetic Textures in Fe_3GaTe_2 Revealed by Lorentz Transmission Electron Microscopy," *2026 MRS Fall Meeting & Exhibit*, Symposium QT01 (QT01.08), Boston, MA, Dec. 2, 2026 (accepted).

WORK EXPERIENCE

Engineer (Full-time), **Logic Technology Development Team** Mar 2023 – Mar 2025
Semiconductor R&D Center, Samsung Electronics, South Korea

1) Back-end-of-line (BEOL) process integration, 1.4 nm GAA logic node (Samsung Foundry SF1.4)

- Integrated BEOL schemes from mask test-structure design (Calibre DRC/LVS) to development-lot evaluation, coordinating process teams.
- Combined inline metrology, electrical testing, and cross-sectional TEM to optimize interconnect stacks for resistance, process stability, and yield.
- Reduced contact resistance (Rc) by ~10% and resolved hard-mask etch drift through over-etch and stackdown-margin optimization, mitigating line-open defects.

2) Backside BEOL route development and integration for backside power delivery (BSPDN)

- Developed a backside BEOL process route and verified integration windows through wafer evaluation and electrical testing.
- Achieved 100% nTSV-to-M1 landing success in an evaluated development test vehicle by resolving backside alignment challenges.

Intern (Full-time), **Logic Technology Development Team** Jul 2022 – Aug 2022
Semiconductor R&D Center, Samsung Electronics

Next-generation Ru BEOL interconnect exploration, 1.0 nm logic node

- Evaluated Ru BEOL interconnect options and integration schemes for future logic nodes.

TECHNICAL SKILLS

- **Electron microscopy:** TEM, LTEM, SEM, FIB
- **Image analysis & computation:** Python, JavaScript, DigitalMicrograph, MuMax3
- **Fabrication:** Glovebox exfoliation and dry transfer, lithography, metal deposition, ion milling, wire bonding
- **Characterization:** MOKE, Raman spectroscopy, VSM, PPMS, AFM, probe station
- **Semiconductor CAD:** Calibre (DRC/LVS), KLayout

AWARDS & HONORS

Quarterly Team Performance Recognition Nov 2023
Logic Technology Development Team, Samsung Electronics

Semester High Honors (Academic Excellence) Fall 2020, Spring 2021, Fall 2021, Spring 2022
College of Engineering, Korea University

Citations for Exemplary Service Aug 2019, Oct 2019
Commissioner and Mobile Corps Commander, Seoul Metropolitan Police Agency

LEADERSHIP & SERVICE

Vice President, Korea University MSE Undergraduate Society (KUMUS) Fall 2021 – Spring 2022

- Organized weekly seminars on faculty research and coordinated industry events.

Vice Captain, FC MSE (department soccer team) Fall 2021 – Spring 2022

Volunteer Tutor, Seomna Child Welfare Center Mar 2019 – Jan 2020

- Mentored underserved elementary and middle school students in math and science (65 hours).

MILITARY SERVICE

Auxiliary Police Officer (Compulsory Service), Seoul Metropolitan Police Agency Oct 2018 – May 2020